

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 12: Aircraft Batteries & Thermal Runaway

Syllabus Source: CAE Oxford (pp. 43-68 (Elec)) | Pass Target: >= 90% AviationExam

1. AIRCRAFT BATTERY CHEMISTRIES: LEAD-ACID VS NICKEL-CADMIUM (NI-CAD)

Aviation storage batteries supply emergency DC electrical power, APU/engine starting current, and stabilize DC bus voltage. Two incompatible electrochemical battery chemistries are used in transport aviation:

Feature	Lead-Acid Battery	Nickel-Cadmium (Ni-Cad) Battery
Electrolyte	Dilute Sulphuric Acid (H ₂ SO ₄) in water (S.G. ~1.280 fully charged).	Potassium Hydroxide (KOH) in water (~30% solution). DOES NOT CHANGE S.G. during charge/discharge!
Cell Voltage	2.0 Volts per cell nominal (12 cells in series = 24V battery).	1.2 Volts per cell nominal (20 cells in series = 24V battery).
Discharge Curve	Voltage drops progressively and steadily throughout discharge.	Maintains virtually FLAT constant voltage output until 90% discharged, then drops precipitously.
State of Charge Check	Measured accurately using a hydrometer to check electrolyte Specific Gravity (S.G. drops as battery discharges).	CANNOT be checked by hydrometer or voltage! (KOH is only a catalytic medium; voltage remains constant). Must undergo controlled workshop discharge test.
Cold Weather Performance	Discharged battery can freeze at -10°C (electrolyte becomes pure water).	Superior cold-weather starting performance; electrolyte freezing point is -60°C.

[!] EXAM TRAP (AVIATIONEXAM) - Thermal Runaway in Nickel-Cadmium (Ni-Cad) Batteries

- Mechanism: Ni-Cad batteries have a **NEGATIVE TEMPERATURE COEFFICIENT OF RESISTANCE!**
 As battery temperature rises, internal electrical resistance **DECREASES**. - The Vicious Cycle: At constant-voltage charging, lower resistance draws **HIGHER** charging current -> generates **MORE** heat -> resistance drops further -> draws even **MORE** current -> violently boils electrolyte, releases explosive hydrogen gas, and causes violent battery explosion/fire! - Protection: Ni-Cad battery compartments **MUST** be equipped with battery temperature sensors and thermal cut-off switches!

2. BATTERY CAPACITY & EMERGENCY STANDBY DURATION

Battery capacity is rated in Ampere-Hours (Ah). A 40 Ah battery can theoretically supply 40 Amperes for 1 hour, or 4 Amperes for 10 hours:

- CS-25 Emergency Duration: In the event of complete primary electrical power generation failure (dual generator loss), aircraft emergency batteries **MUST** supply essential flight deck instruments, emergency lighting, and VHF comms for **AT LEAST 30 MINUTES**.
- Separation of Maintenance Shops: Lead-acid and Ni-Cad batteries must be serviced in **COMPLETELY SEPARATE** battery servicing rooms. Acid fumes from lead-acid neutralize alkaline KOH electrolyte in Ni-Cad cells, destroying both batteries!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Battery State of Charge & Thermal Runaway**

1. Lead-Acid: State of charge is proportional to electrolyte specific gravity (1.280 fully charged down to 1.150 discharged) and open-circuit voltage. 2. Ni-Cad: State of charge CANNOT be measured by specific gravity (KOH electrolyte does not change density). Measured only by complete discharge/recharge bench test. 3. Thermal Runaway: Ni-Cad batteries have negative temperature coefficient. Charging at constant voltage causes runaway. Ni-Cad batteries MUST be charged with CONSTANT CURRENT.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the fundamental electro-chemical mechanism that triggers thermal runaway in an aircraft Nickel-Cadmium battery?**

- [A]: High internal resistance at freezing temperatures
- [B]: A negative temperature coefficient of resistance, where increasing internal temperature causes internal resistance to drop, drawing exponentially higher charging current under constant voltage
- [C]: Acid vaporization
- [D]: Over-dilution of sulfuric acid

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Ni-Cad cells have a negative temperature coefficient. As they heat up, internal resistance drops, causing the battery to draw more current from the constant-voltage bus, producing more heat in an uncontrollable thermal runaway cycle.

Q2: How can the state of charge of an aircraft lead-acid battery be determined accurately in a maintenance facility?

- [A]: By measuring the specific gravity of the electrolyte with a hydrometer
- [B]: By shaking the battery
- [C]: By weighing the battery plates
- [D]: By checking battery temperature

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. As a lead-acid battery discharges, sulfuric acid is consumed to form lead sulfate and water, dropping electrolyte specific gravity from 1.280 to 1.150. S.G. directly indicates state of charge.

Q3: Under CS-25 certification requirements, for what minimum time period must emergency aircraft batteries supply power to essential instruments and communication equipment?

- [A]: 10 minutes
- [B]: 15 minutes
- [C]: 30 minutes
- [D]: 60 minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Transport category aircraft batteries must provide at least 30 minutes of emergency power to essential flight instruments, standby attitude indicator, and communication equipment.